

AION Device Fabric

Games and Education Proof-of-Concept Supplement

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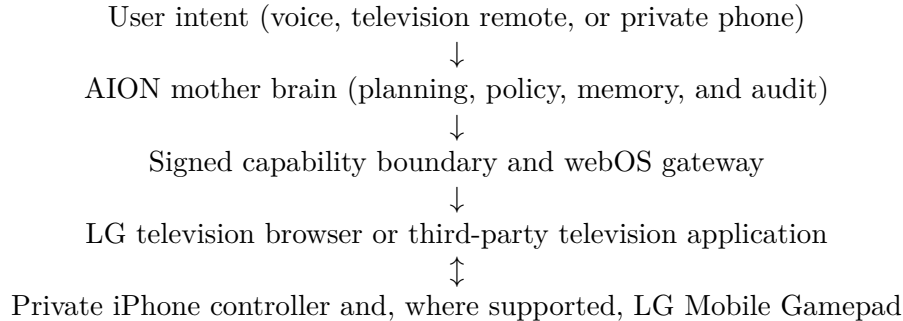
Fabric implementation version 0.14.0

Abstract

This supplement records two application proofs implemented on top of the AION Device Fabric: a governed cloud-gaming surface and an interactive Spanish learning centre. Neither proof requires additional television hardware. The existing LG webOS television is represented by a permissioned gateway, the laptop operates as the mother node, and an iPhone provides a private control surface. The proofs demonstrate that a restricted television can become a room-scale surface for composed intelligence, external services, local state, and cross-device interaction.

1 Common Fabric Substrate

Both applications reuse the same Device Fabric rather than installing separate systems. The relevant topology is:



The mother brain retains the intelligence, policies, service state, and audit evidence. The television receives a temporary, token-protected presentation surface. The phone receives a separate private companion token and CSRF secret. Television display state therefore does not contain the webOS pairing key, COMDEX credentials, payment data, or private service credentials.

2 AION Games

2.1 Objective and demonstrated result

AION Games composes an existing television, the NVIDIA GeForce NOW web client, the AION webOS gateway, and a phone-based controller into a low-hardware cloud gaming experience. A command such as “*AION, open games*” causes AION to open the allowlisted GeForce NOW surface at <https://play.geforcenow.com/>. Game rendering and streaming remain an NVIDIA service; AION supplies discovery, navigation, private text delivery, controller composition, and governance.

The proof established that a compatible household television can expose a large cloud-game catalogue without a console. The user still requires a valid NVIDIA account, any applicable subscription, a supported game entitlement, sufficient network quality, and a controller path accepted by the television and streaming service.

2.2 Execution flow

1. A voice or phone command is parsed as the governed `open_games` intent.
2. The mother selects the enrolled LG gateway and verifies that the games capability is present.
3. The webOS adapter opens the fixed GeForce NOW URL in the television browser and records a verified action receipt.
4. The private phone companion supplies directional navigation, OK, back, pointer movement, pointer click, and game search controls.
5. If typed sign-in is necessary, the user enters the username or password on the phone. AION sends that value once to the currently focused television field. The value is cleared from the form and is not placed in the Fabric audit ledger.
6. CAPTCHA, two-factor authentication, legal consent, account selection, purchases, and the final sign-in decision remain explicit user acts.
7. Once a stream begins, gameplay input can transfer to the LG Mobile Gamepad or another controller exposed by the platform.

2.3 Control and trust boundaries

The GeForce NOW URL is allowlisted; arbitrary URL execution is not granted by the games command. The TV touchpad exists because some third-party pages require pointer input, but it does not authorize AION to accept contracts, purchase subscriptions, defeat access controls, or infer consent. Account secrets are intentionally excluded from normal Fabric state, telemetry, screen projection, and audit receipts.

The phone controller is delivered on the private LAN through a rotating session URL. Each state-changing request carries a CSRF value, is rate limited, returns to the mother brain, and passes through the same signed TV policy used by voice commands. A phone waking from sleep performs bounded retries and a state refresh; loss of a transient browser connection therefore does not create a new television enrollment.

2.4 Implemented components

Component	Responsibility
<code>runtime.py</code>	Governed companion commands, games capability routing, and signed action policy.
<code>webos_gateway.py</code>	LG pairing, application and browser launch, remote-key delivery, pointer input, and verified receipts.
<code>companion.py</code>	Private remote, touchpad, one-shot credential delivery, game search input, reconnect behaviour, and CSRF enforcement.
<code>tv_canvas.py</code>	Room-scale AION Games status and phone-pairing projection.

2.5 Proof boundary and production work

The proof demonstrates composition, not a replacement for NVIDIA, LG, or game licensing. Browser compatibility, codec support, controller exposure, network latency, service-region restrictions, account status, and vendor UI changes remain external dependencies. A production release requires TLS on the phone surface, durable device discovery, controller compatibility testing across LG and Samsung models, accessible sign-in handoff, service health monitoring, and clear subscription and data-use disclosures.

3 AION Education: Spanish Learning Centre

3.1 Objective and demonstrated result

The Spanish Learning Centre is a dedicated television application surface for children approximately six to eight years old who speak English and are learning Spanish. It combines pictures, spoken Spanish, multiple-choice interaction, immediate correction, gamified progress, adaptive repetition, and short conversations. The proof demonstrates that the same Fabric used to control a restricted television can also host a stateful, personalised learning experience.

3.2 Learning model

The current curriculum contains foundational greetings, kind words, food and drink, home, school, people, and feelings. Each word card contains a canonical identifier, Spanish form, English meaning, category, and visual cue. For example, *manzana* is presented together with an apple image cue. Every fourth round is a short conversation challenge rather than a vocabulary card.

Two anonymous local learner profiles are provided. Each profile maintains:

- attempts and correct answers;
- stars and current answer streak;
- per-card mastery values;
- a 50-question session score; and
- the current lesson phase and round.

Cards with lower mastery are selected earlier. A correct answer raises card mastery and an incorrect answer reduces it slightly, causing weak material to return sooner. This is deliberately a small deterministic adaptive loop for the proof; it does not claim to be a validated educational assessment model.

3.3 Television interaction

The dedicated application is served at the temporary Fabric path `/tv/<session>/education`. It exposes large focusable answer controls, visual cues, score and question progress, and controller connection status. Input is accepted from legacy webOS remote key codes, normal browser arrow and Enter events, or the browser Gamepad API when LG exposes a compatible gamepad.

After selection, the interface:

1. disables further answers for the current round;
2. shows a green tick for a correct answer or a red cross for an incorrect answer;
3. always identifies the correct answer;

4. updates the score, stars, streak, attempts, and mastery;
5. speaks the feedback; and
6. advances automatically after speech finishes, with a bounded timeout fallback.

Automatic advancement includes the expected round number. The mother rejects a duplicate or stale advance from a second browser or reconnecting controller, preventing accidental skipped questions. Periodic state refreshes also preserve the currently focused answer instead of rebuilding focus every cycle.

3.4 Natural speech pipeline

The television does not need native speech synthesis. It requests the current bounded lesson phrase from the mother at `/education/audio`. The mother applies the following provider chain:

1. configured ElevenLabs `eleven_multilingual_v2` natural voice;
2. configured OpenAI `gpt-4o-mini-tts` natural voice;
3. local macOS Spanish speech as an offline fallback.

Cloud requests contain only the fixed lesson phrase, limited to 240 characters. They do not contain a child’s name, recorded voice, microphone audio, profile history, answer history, or household identity. Returned audio is limited to 2 MB and cached only in process memory with a bounded 96-entry cache. The television interface discloses that the natural learning voice is AI-generated. The audio control also permits an explicit *Hear AI voice again* action.

3.5 State, actions, and security

Persistent learning state is stored beneath the Fabric runtime directory in `education/learning_centre.json`. State writes use a temporary file followed by an atomic replacement. The public session projection removes the hidden correct-answer index before it is sent to a client.

The television action endpoint accepts only `start`, `answer`, `next`, and `repeat`. Requests must include the independent `X-AION-Education` session secret and remain below the request-size limit. The phone uses the separate companion token and CSRF secret. Education actions are recorded in the Fabric audit chain without recording private speech or credentials.

The child-safety posture currently enforces no child account, no advertising, no open-web link, no purchase path, and no retained raw voice. Learner labels are abstract local profiles rather than legal identities. The mother microphone can remain physically closed while the lesson’s output voice continues to operate.

3.6 Implemented components

3.7 Verified proof status

At the time of this record, the implementation has demonstrated:

- discovery, pairing, and verified control of the LG webOS television;
- launch of a dedicated learning page on the television;
- persistent independent progress for two anonymous learners;
- remote, phone, and compatible gamepad answer navigation;
- picture-supported vocabulary and conversation rounds;

Component	Responsibility
<code>education.py</code>	Curriculum, anonymous profiles, mastery selection, 50-question sessions, scoring, feedback, and atomic persistence.
<code>tv_canvas.py</code>	Dedicated visual learning application, remote and gamepad mapping, protected actions, natural speech delivery, focus retention, and automatic progression.
<code>companion.py</code>	Phone-based answers, lesson controls, private session state, and wake/reconnect behaviour.
<code>runtime.py</code>	Education capability registration and governed routing from voice or phone intent.
<code>cli.py</code>	Live service composition, television lesson handler, audit events, and stale-advance protection.

- correct/incorrect visual feedback and explicit correction;
- automatic, duplicate-safe progression;
- natural Spanish speech with provider fallback;
- phone reconnection after browser sleep; and
- 127 passing automated Fabric tests with a valid audit chain.

3.8 Remaining production work

The next education phase should replace pictographic cues with a curated, licensed illustration library; expand the curriculum and sentence model; introduce age-appropriate difficulty bands; test pronunciation and regional Spanish variants with educators; add parent-visible learning summaries; measure retention across days; add accessibility controls; prefetch speech to remove first-play latency; and conduct child-safety, privacy, and educational efficacy review. Natural speech providers also require explicit account, quota, disclosure, retention, and offline-mode configuration in a consumer installer.

4 Conclusion

The two proofs show the same architectural result from different directions. AION Games composes a television, cloud service, phone, and controller into a new entertainment surface. AION Education composes the television, local learner state, visual interaction, adaptive sequencing, and natural speech into a new household learning surface. The television is therefore not the intelligence host by necessity; it is a governed room-scale endpoint of the Fabric, while the mother brain supplies intelligence, memory, permissions, recovery, and evidence.